MIL-H-60840(MU)
22 May 1968
SUPERSEDING
FA-PD-MI-2543, Rev.0
20 September 1962

MILITARY SPECIFICATION
HOLDER, STRIP, ASSEMBLY, METAL PARTS

1. SCOPE

1.1 This specification covers Holder, Strip, Assemblies for use in the M8, M9, M10, XM12 or M19 Aircraft Ejection Seat Catapults.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to this extent specified herein.

SPECIFICATIONS

Military
MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-P-10025 - Packaging, Packing and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for

STANDARDS

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Card

DRAWINGS

F8596896 - Holder, Strip, M9 Assembly
F8596934 - Holder, Strip, Subassembly
F8596959 - Holder, Strip, Subassembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1377

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2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B 46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N. Y. 10016)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.- When protective finish is required by component details, finish requirements shall be as required on the drawings and referenced specification and shall apply to all surfaces of the assembly regardless of contour configuration.

3.3 Assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of Drawing F8596896 or F8596934 or F8596959.

3.4 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following.

3.4.1 Metal defects.- All metal parts shall be free from cracks, splits, folds, or other metal defects.

3.4.2 Surface finish.- The requirements for surface finish are as detailed on the applicable drawings and referenced standard.

3.4.3 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.4.4 Foreign matter.- No part or assembly shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless

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4.1 (Cont'd)

disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 25 strip holder assemblies, 50 strips for hardness test and 10 sets of unassembled components. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all the requirements of the drawings and specifications, at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to meet the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be presented in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 holder strips, maximum, exclusive of test samples, provided the homogeneity of manufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

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4.3.2 Examination.- One hundred percent examination shall be performed for critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects using the sampling plans and criteria in Tables I and II. For an individual defect (single characteristic), the criteria in Table I shall be used. For total defects within a major or minor group, the criteria in Table II shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The roughness comparison specimens prescribed by ASA B46.1 shall be used as a basis of comparison for surface roughness determination.

Table I

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 -22000	300	4	5	5	6

Table II

	Total Allowable Defects Per Sample Size			
Sample Sizes for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14
Sample Sizes for Minor Defects	110	150	225	300
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

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4.3.2.1 Holder, Strip, M9, Assembly, Drawing F8596896.
Holder, Strip, Subassembly, Drawing F8596934.
Holder, Strip, Subassembly, Drawing F8596959.

	Defect	Method of Inspection
CRITICAL		
1.	Location and number of holes	Gage
2.	Cable terminals missing or insecure	Manual & Visual
3.	Cable guide missing or insecure	Manual & Visual
4.	Metal defective	Visual
5.	Broken wire strands in cable	Visual
6.	Orientation of swaged terminals	Visual
7.	Orientation of cut-outs to cable guide	Visual
MAJOR		
101.	Diameter of holes	Gage
102.	Width of strip	Gage
103.	Length of strip	Gage
104.	Extended length of cable and strip	Gage
105.	Thickness of strip	Gage
106.	Location of cable guide	Gage
107.	Dimensions of notches incorrect	Gage
108.	Location of notches	Gage
109.	Cable guide weldment incomplete or incorrect	Visual
110.	Number of notches incorrect	Visual
MINOR		
201.	Radii missing	Visual
202.	Plating inadequate or improper	Visual
203.	Surface finish improper	Visual
204.	Marking missing, incorrect or illegible	Visual
205.	Burr	Visual
206.	Foreign matter	Visual

4.3.3 Inspection.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Hardness.- A sample of 50 strips shall be selected at random from the lot and tested, before plating, for hardness in the area as required by the drawing. Failure of any strip to meet the requirement shall be cause for rejection of the lot.

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4.3.3.2 Protective finish.- The number of strips to be used for the protective finish (plating) tests shall be in accordance with the applicable specification. Failure of any sample to comply with the requirements of the specification shall be cause for rejection of the lot.

4.3.3.3 Cable, terminal and guide security.- Each assembly shall be subjected to a pull test in accordance with 4.4.3. Failure of the cable, cable guide weldment or the cable terminal swages to withstand the pull test without deformation, movement, disorientation, loosening or breakage of part or connection, except for inward or outward movement of cable wires not greater than .005 of an inch, shall be cause for rejection. Assemblies which exhibit .005 wire movement when subjected to the pull test shall be segregated from the lot until completion of the destructive test specified in 4.4.3.1.

4.3.3.3.1 Cable, terminal and guide assembly - destructive test.- A sample of 25 assemblies shall be selected from the entire lot. Sample shall be made up primarily of assemblies which exhibit evidence of center wire movement when quantities permit, otherwise, assemblies which perform normally in the pull test shall be used to make up the required sample size. The 25 assemblies shall then be tested in accordance with 4.4.3.1. Failure to meet the requirements specified on the applicable drawing shall be cause for rejection of the lot.

4.3.3.4 Ball fitting.- Each strip holder assembly shall be tested after swaging to determine that the ball fittings are free from all discontinuities; such as cracks, splits and folds. Failure of any strip assembly to comply with the requirements of 3.4.1 shall be cause for rejection of the assembly.

4.4 Test methods and procedures.

4.4.1 Hardness test.- Prior to plating, a hardness testing machine shall be used to determine hardness.

4.4.2 Protective finish (plating) test.- The method and procedure for performing these tests shall be in accordance with the applicable specification referenced.

4.4.3 Cable, terminal swaging and cable guide weldment test.- The cable assembly shall be tested by applying 2000 (+ 50) pound tensile load for 5 (+ 1) seconds. Apply the load from the "ball end" terminal to the "shank end" terminal and against cable guide simultaneously in the direction designated on the applicable drawing.

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4.4.3.1 Cable, terminal swaging and cable weldment - destruction test.- The test shall be performed in accordance with 4.4.3 except that the pull shall be continued until destruction. The force required to break the cable shall be recorded.

4.4.3.2 Ball fitting test.- The following Magnaflux Corporation Zyglo Penetrex Method, or equal, shall be used to determine that the ball fittings (ball and shank) of each strip holder assembly are free from discontinuities, such as cracks, splits and folds after swaging:

a. One-half hour time lapse following the application of ZL-22 Penetrant.

b. One minute emulsification time using ZE-3 Emulsifier.

c. Remove penetrant and emulsifier with forceful water stream (30-pound pressure). Avoid overwashing.

d. Dry in a hot-air drier operating at 225°F. and allow to cool.

e. Apply ZP-4 Developer and examine ball fittings under a light with a wave length of 3650 to 4000 Angstrom units.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-10025, or as required by contract.

5.2 Data cards.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 Supersession data.- This specification includes the requirements of Purchase Description FA-PD-MI-2543, Rev. 0 dated 20 September 1962.

Custodian:

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Preparing activity:

Army - MU

Project No. 1377-A215

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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